

MADANI SAI VARDHAN

madanisaivardhan@gmail.com | +91 6302671238 | linkedin/sai-varadhan-madani-25488330a | github/Sai-Vardhan12 | Nizamabad, Telangana

PROFESSIONAL SUMMARY

CSE (Data Science) graduate with hands-on experience in Machine Learning, Deep Learning, Computer Vision, and Data Analytics. Proficient in Python, SQL, and AI/ML frameworks. Completed two Data Science internships — large-scale dental image annotation at Solaiera and Data Science & Analytics at Vortex Technologies. Built 4+ end-to-end AI/ML projects. Eager to build intelligent, data-driven solutions.

TECHNICAL SKILLS

Programming & Query: Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), Java, C, SQL (MySQL, PostgreSQL)

AI / ML: Machine Learning, Deep Learning, Federated Learning, Computer Vision, NLP, Generative AI, CNN, SHAP

Tools & Platforms: Power BI, Tableau, Roboflow, Streamlit, Jupyter Notebook, Google Colab, Git, GitHub, VS Code, Excel

Concepts: Data Analytics, Data Science, Privacy-Preserving ML, Explainable AI, Image Annotation, EDA

INTERNSHIP EXPERIENCE

Data Science Intern – Solaiera (solaiera.ai)

Feb 2026 – May 2026

- Annotated 400+ dental medical images using bounding box & polygon segmentation on Roboflow; conducted QA checks for missing labels and misclassifications to build ML-ready datasets.

Data Science & Analytics Intern – Vortex Technologies

Jul 2026

- Completed a 1-month internship demonstrating exceptional technical skills and dedication; made significant contributions to core Data Science & Analytics projects.

PROJECTS

AI Security Log Analyzer | *Python, XGBoost, SHAP, Streamlit*

- Built a multi-source AI security analytics platform with XGBoost intrusion detection, Isolation Forest anomaly detection, SHAP explainability, and a real-time Streamlit dashboard with alert prioritization.

Privacy-Preserving & Explainable Federated Learning | *Python, Federated Learning, SHAP,*

- Designed a federated learning framework for multi-disease risk prediction using wearable sensor data; integrated differential privacy and SHAP-based explainability to ensure data confidentiality and model transparency across distributed clients.

HyperVision – Hyperspectral Image Detection | *CNN, PCA, SHAP, Python*

- Developed a CNN-based hyperspectral object detection system with PCA dimensionality reduction, drift detection, adaptive learning, and SHAP explainability.

MediBot – AI Medical Assistant Chatbot | *Python, NLP, Healthcare AI*

- Built a healthcare chatbot for symptom tracking, medication/appointment management, and conversational health guidance with urgent-concern flagging logic.

EDUCATION

B.Tech – CSE (Data Science), Vardhaman College of Engineering, Hyderabad

2023 – 2027 | CGPA: 8.27

Intermediate (MPC) – S.R. Junior College

2021 – 2023 | 96.4%

High School – Vijaya High School

2021 | CGPA: 10.0

CERTIFICATIONS & TRAINING

NPTEL: Data Analytics with Python

Oracle: Cloud Infrastructure (OCI), Oracle DBA 21c

Others: AI for Beginners, Cybersecurity Awareness (HP), LinguaSkill Business English (Cambridge)

INTERESTS & HOBBIES

- Competitive Programming
- AI & Emerging Technologies
- Photography
- Cricket

DECLARATION

I hereby declare that all the information provided above is true and accurate to the best of my knowledge and belief. I take full responsibility for the correctness of the above-mentioned details.

Madani Sai Vardhan